David Lydston

Principal Product Manager | Data Platforms, Metric Governance & Experimentation

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Profile

Product leader building governed data platforms, common metric definitions, and reliable measurement for **30+ products and 3,000+ experiments monthly**. Hands-on background in customer analytics and clinical data management. Set vision for a **24-person team**; directly managed four individual contributors.

Core expertise

Product strategy and roadmaps · Metric governance and catalog adoption · Metadata and schema requirements · Data quality and validation · Causal data analysis · Structured Query Language (SQL), SQL Server, Apache Spark

Professional experience

Principal Product Manager | Microsoft | Central Data & Experimentation | Aug 2024 – Present

Lead shared experiment measurement for Microsoft 365 and Copilot, Microsoft's generative artificial intelligence assistant, so product teams evaluate changes using consistent definitions.

- Required metrics to use Microsoft's **Unified Data Model**, a governed shared data model; championed adding experiment-assignment data for **30+ products** to connect tests with common user, organization, and device metrics.
- Standardized measurement across **hundreds of features** by moving key metrics to team-owned scorecards; clarified ownership, readiness, and service targets across partner teams.
- Cut results delivery from **three days to 36 hours** after daily data collection and reduced compute consumption **more than 75%** through leaner scorecards and faster metric processing.
- Protected sensitive customer data by rejecting publication on unsecured scorecards during platform transitions.

Principal Product Manager | Microsoft | Office Product Group | Nov 2022 – Aug 2024

Led experimentation infrastructure for Office products, helping teams evaluate growth initiatives through reliable measurement.

- Co-authored the **fiscal year 2025 experimentation strategy**, adopted by senior leadership as the organization's vision.
- Prioritized Copilot usage scorecards and connected data-quality alerts to engineering bug triage, helping teams identify and address unreliable results.

Principal Product Manager | Microsoft | Analysis & Experimentation / Bing | Sep 2020 – Nov 2022

Led enterprise experimentation initiatives spanning Bing release decisions, cloud compliance, and compute efficiency.

- Delivered **$7 million in scorecard compute-cost savings with Bing**; co-led optimizations that shortened Bing's release-decision cycle **from six days to two**.
- Prioritized five core Microsoft Azure platform certifications and production readiness, enabling adoption by Dynamics 365 and the Azure Portal.
- Negotiated Bing cost sharing for migration from Microsoft's internal big-data processing platform to **Apache Spark**.

Senior Program Manager | Microsoft | Analysis & Experimentation / Bing | Aug 2018 – Sep 2020

Improved measurement quality, targeting, and processing reliability for Microsoft's experimentation platform and Bing.

- Improved intraday alert accuracy **50%** with bot-tolerant, user-based metrics, virtually eliminating false positives.
- Authored audience-targeting filters for customer segments and migrated the real-time experiment-assignment service under a fixed deadline.

Senior Program Manager | Microsoft | Marketing & Store Analytics | Feb 2016 – Aug 2018

Delivered digital analytics for Microsoft Store, Windows, and news properties, improving shared data definitions, access, and operating efficiency.

- Contributed cohort and filter requirements covering identifier semantics, schema compatibility, time windows, usage metadata, and query performance for a shared analytics platform.
- Retired **28 administrator accounts** and ended account sharing; reduced Adobe Analytics sampling volume **50%**, saving **more than $60,000 monthly**.

Earlier experience

Data Scientist (Client Services & Strategy) | HelloWorld (formerly ePrize) | May 2012 – Jan 2016

Applied customer data science to Microsoft's Xbox Live Rewards loyalty program, serving **3 million members**; translated behavioral data into marketing, personalization, and program-design decisions.

- Cleaned granular transaction and behavioral events into reusable machine-learning datasets; built classification, recommendation, and fraud-detection models.
- Used machine learning to inform **MyVIP loyalty tiers and qualifying thresholds**, connecting member behavior with program incentives.
- Evaluated MyVIP against stratified control groups, measuring **300% more points earned** in fiscal year 2015 versus controls.

Database and Systems Engineer | Apollo Data Technologies | Microsoft contract: Xbox Live Rewards | Aug 2010 – May 2012

Built the processing, reporting, and analysis foundation for a customer loyalty program, balancing performance, reliability, and hardware constraints.

- Helped **launch Xbox Live Rewards in December 2010**; designed infrastructure for a **five-terabyte data-processing platform**.
- Reduced processing time **30%** and data-layer storage **2.5 terabytes** through code optimization, compression, and index tuning; trained developers on performance tradeoffs.

Database and Systems Engineer | Apollo Data Technologies | Microsoft contract: Global Marketing Operations | Jul 2008 – Aug 2010

Engineered database infrastructure for global marketing applications, including capacity planning, monitoring, standardized deployment, and operational handoff.

- Selected infrastructure within a **$1 million hardware budget** during threefold growth, supporting hundreds of marketing applications and thousands of databases.
- Implemented monitoring for **up to 500 servers**; standardized SQL Server deployment and trained onshore and offshore teams.

Senior Database Administrator | Bio-Rad Laboratories | Aug 2005 – Jul 2008

Combined clinical data management, database administration, and biostatistics for a medical diagnostics manufacturer.

- Replaced per-trial spreadsheets with a **centralized clinical-trial data management system**; automated validation saved weeks of manual work. Recalled in 2013 to extend the system.
- Used **Statistical Analysis System (SAS)** software for instrument precision and clinical analyses supporting **U.S. Food and Drug Administration (FDA) submissions**, including mixed-effects and survival models.

BioAnalyst | Poniard Pharmaceuticals | Nov 2004 – Aug 2005

Built clinical data capture and analysis capabilities for a pivotal Phase III oncology trial.

- Specified, designed, tested, and implemented a medical-chart information system; partnered with a contract research organization on trial data-quality standards.
- Performed survival analysis in SAS and produced **hundreds of clinical result tables** for FDA submission.

Systems Analyst III | University of Miami | Aug 1999 – Nov 2004

Led development of SQL Server applications for clinical research operations and financial planning.

- Delivered participant-tracking systems and handheld interview data capture for multi-state and multinational clinical research.
- Led an application forecasting federal grant expenditures against a **$25 million departmental budget**.

Education

Master of Science, Information Systems Management — Seattle Pacific University, 2007
Master of Arts, Behavioral Psychology — San Diego State University, 1999
Bachelor of Arts, Psychology — Western Washington University, 1997